Insurance Fraud Detection Using Network Analytics

Project Type: Data Analytics & Business Intelligence
Tool Used: Microsoft Power BI
Dataset Size: 5,000 Insurance Claims
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Declaration

I hereby declare that the project titled **"Insurance Fraud Detection Using Network Analytics"** is my original work and has been carried out by me using Microsoft Power BI. This work has not been submitted previously for the award of any degree, diploma, or certification at any institution. All sources of data and references used in this project have been duly acknowledged.

Name: ***Vijay Kumar Naik***
Date: 06-02-2026

Acknowledgment

I would like to express my sincere gratitude to my mentors, faculty members, and peers for their guidance and support throughout the completion of this project. I also acknowledge the use of simulated insurance data and Microsoft Power BI for enabling effective analysis and visualization. Their contributions have been invaluable in the successful completion of this project.

Problem Statement

Insurance fraud results in significant financial losses and often involves coordinated activities among multiple entities such as claimants, providers, and agents. Traditional rule-based fraud detection systems focus on individual claims and fail to capture relationship-driven fraud patterns. The challenge is to identify high-risk entities and organized fraud networks using explainable analytics techniques.

Project Objectives

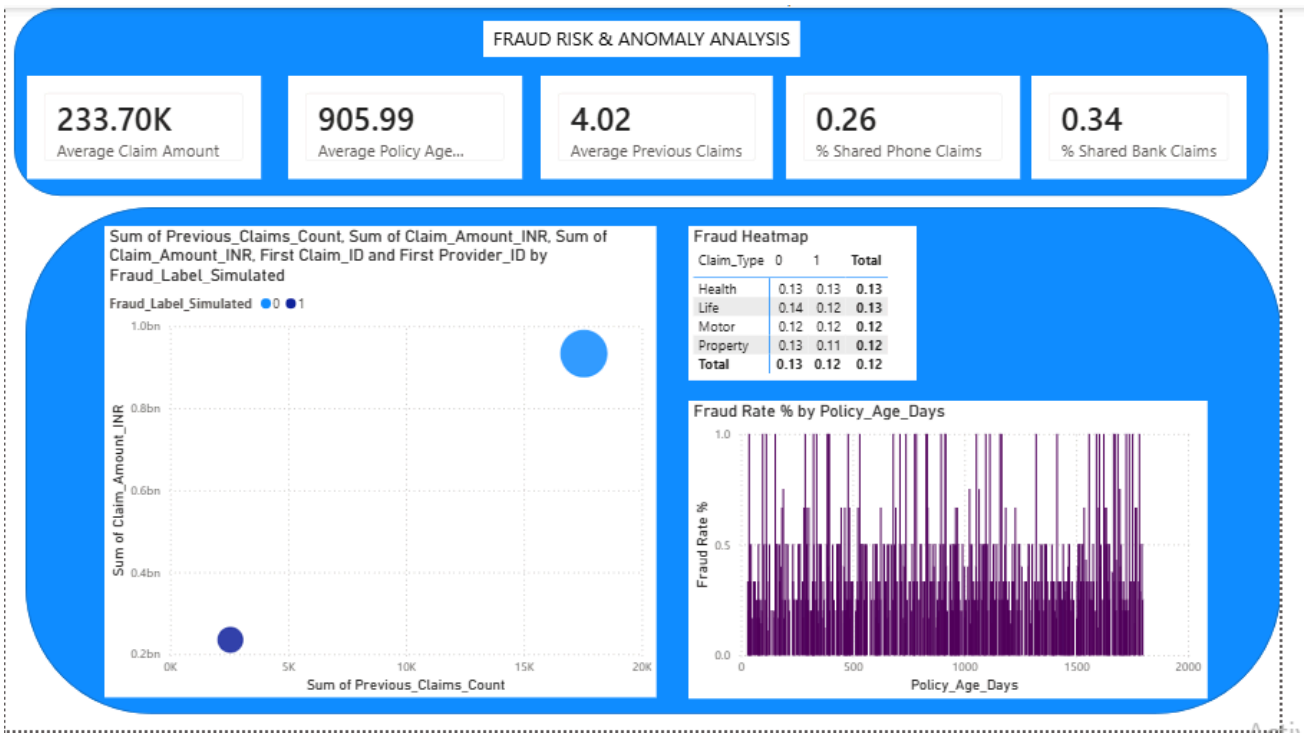
- To analyze insurance claims data using network-oriented analytics principles.
- To identify high-risk providers and agents acting as fraud hubs.
- To design interactive Power BI dashboards for fraud monitoring and investigation.
- To prioritize fraud investigations using scoring and funnel-based analysis.
- To deliver an explainable and scalable fraud analytics solution.

Executive Summary

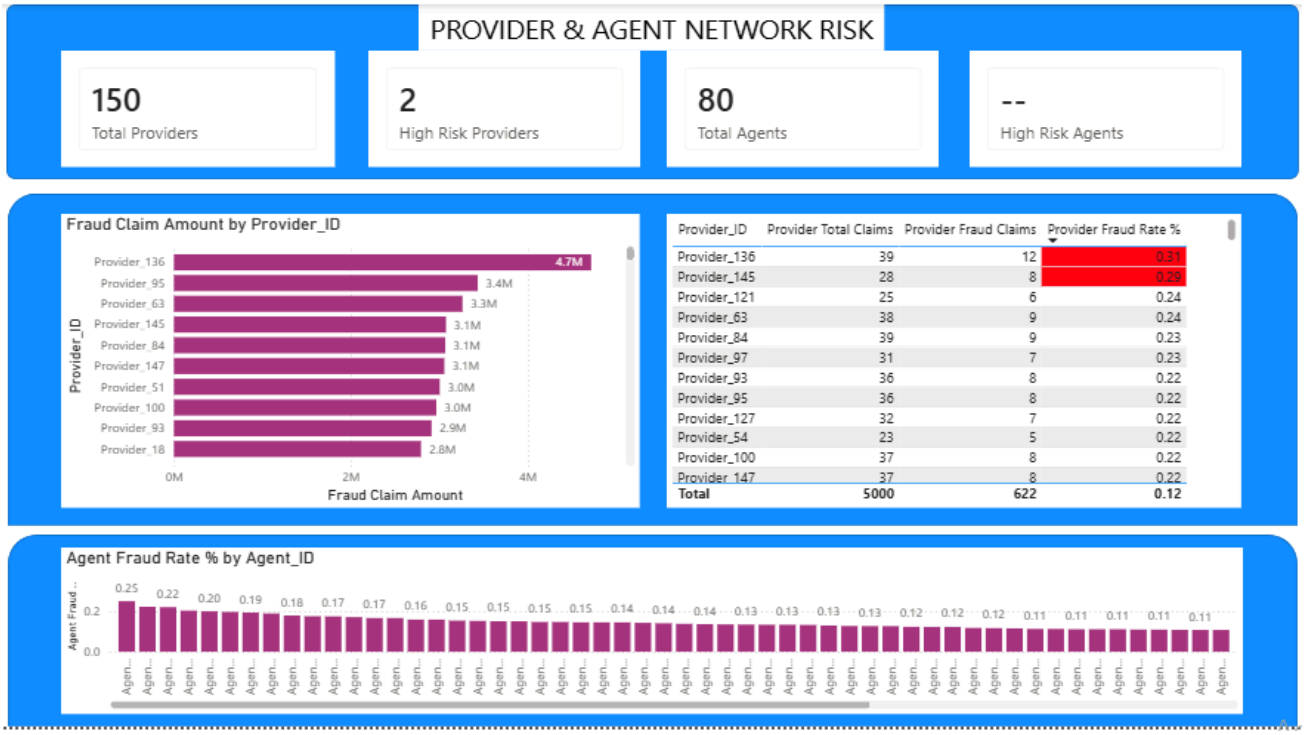
Insurance fraud causes significant financial loss and often involves coordinated behavior across claimants, agents, and providers. This project applies network-based analytics using Power BI to identify fraud patterns, high-risk entities, and investigation priorities through relationship-driven indicators.

Executive Fraud Overview Dashboard

Fraud Risk & Anomaly Analysis Dashboard

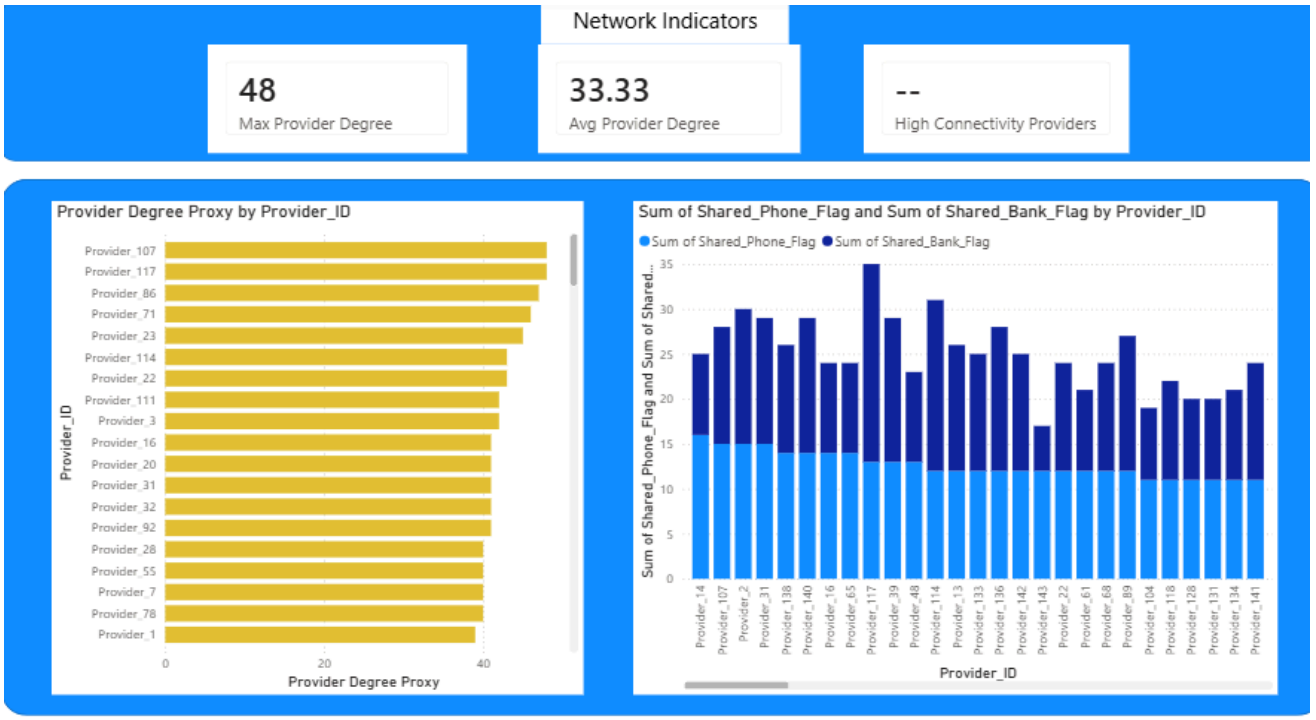


Provider & Agent Network Risk Dashboard

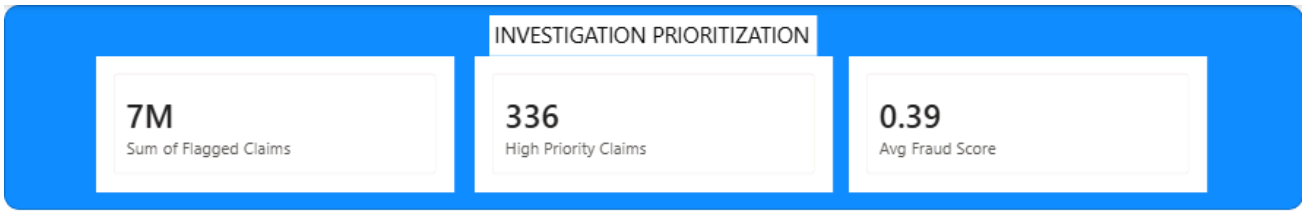


Provider_ID	Provider Total Claims	Provider Fraud Claims	Provider Fraud Rate %
Provider_136	39	12	0.31
Provider_145	28	8	0.29
Provider_121	25	6	0.24
Provider_63	38	9	0.24
Provider_84	39	9	0.23
Provider_97	31	7	0.23
Provider_93	36	8	0.22
Provider_95	36	8	0.22
Provider_127	32	7	0.22
Provider_54	23	5	0.22
Provider_100	37	8	0.22
Provider_147	37	8	0.22
Total	**5000**	**622**	**0.12**

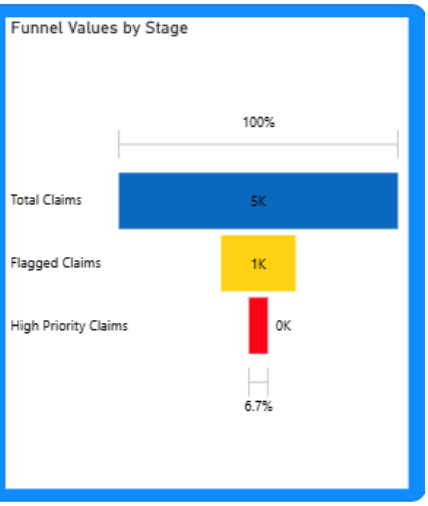
Network Indicators (Centrality Proxy) Dashboard



Investigation Prioritization Dashboard



Claim_ID	Claimant_ID	Provider_ID	Agent_ID	Sum of Claim_Amount_INR	Sum of Fraud Score	High Priority Investigation
CLM004188	CUST01252	Provider_17	Agent_47	449990	0.73	YES
CLM000683	CUST01560	Provider_68	Agent_67	449943	0.67	YES
CLM001201	CUST01527	Provider_81	Agent_58	449756	0.69	YES
CLM000058	CUST00942	Provider_40	Agent_56	449481	0.73	YES
CLM002152	CUST00242	Provider_117	Agent_8	449338	0.79	YES
CLM004813	CUST01606	Provider_149	Agent_49	449319	0.69	YES
CLM000742	CUST00248	Provider_14	Agent_26	448974	0.73	YES
CLM000720	CUST00573	Provider_136	Agent_79	448588	0.83	YES
CLM002955	CUST01182	Provider_106	Agent_16	448456	0.71	YES
CLM003015	CUST00773	Provider_54	Agent_16	447804	0.73	YES
CLM004591	CUST00530	Provider_4	Agent_30	447506	0.73	YES
CLM002696	CUST00052	Provider_132	Agent_23	447203	0.69	YES
CLM001125	CUST00063	Provider_121	Agent_78	446659	0.71	YES
CLM003287	CUST01006	Provider_127	Agent_7	446636	0.71	YES
CLM001969	CUST01617	Provider_96	Agent_12	446548	0.73	YES
CLM003259	CUST00397	Provider_12	Agent_40	445771	0.91	YES
CLM000943	CUST01149	Provider_3	Agent_20	445563	0.69	YES
CLM001585	CUST01207	Provider_138	Agent_24	445249	0.73	YES
CLM004597	CUST01097	Provider_2	Agent_39	444984	0.83	YES
CLM004108	CUST01635	Provider_70	Agent_69	444877	0.69	YES
Total				**110747043**	**241.56**	



Methodology & Data Model

A star-schema data model was implemented with insurance claims as the central fact table. DAX measures were designed to calculate fraud rates, provider and agent risk, network degree proxies, and investigation scores. Network behavior was approximated using claim volume, shared attributes, and percentile-based thresholds.

Business Impact

The dashboards enable early detection of organized fraud, efficient investigation prioritization, and executive-level monitoring. By identifying network hubs and high-risk entities, insurance companies can reduce financial leakage and optimize SIU resources.

Conclusion

This project demonstrates how network analytics concepts can be effectively implemented in Power BI without specialized graph databases. The solution is scalable, explainable, and aligned with real-world fraud investigation workflows.